

(Quiz) Question-1

Driver Code	Output
<pre>cart1= Foodcart() food1=Food('Burger','Chillox',180) food2=Food('Cake','Mr.Baker',1000) print('=====') food1.detail() print('=====') cart1.addFood(food1,food2) print('=====') cart1.setCartName('Foodpanda') cart1.addFood(food1,food2) cart1.addFood(Food('Steak','Woodhouse Grill',2000)) print('=====') cart1.printCartDetails() print('=====') cart1.removeFood('French Fry') print('=====') cart1.removeFood('Steak') print('=====') cart1.printCartDetails()</pre>	<pre>===== Food Info: Burger, Food shop: Chillox, Food price: 180 ===== Please set the cart name first. ===== ===== Details of Foodpanda: Total Items: 3 Food Info: Burger, Food shop: Chillox, Food price:180 Food Info: Cake, Food shop: Mr.Baker, Food price:1000 Food Info: Steak, Food shop: woodhouse Grill, Food price:2000 ===== French Fry not found! ===== Steak removed from cart! ===== Details of Foodpanda: Total Items: 2 Food info: Burger, Food shop: Chillox, Food price:180 Food info: Cake, Food shop: Mr.Baker, Food price:1000</pre>

Question-2

Driver Code		Output	
team= Team('Bangladesh')	team.addPlayer(p4,p5)	=====	=====
p1= Player('Mahmudullah', 'Batsman', 30)	print('=====')	Mahmudullah added in team	=====
print('=====')	team.showPlayers()	=====	Mushfiquir Rahim added in team
team.addPlayer(p1)		Batsman:	Taskin Ahmed added in team
print('=====')		Player name: Mahmudullah Jersey No: 30	=====
team.showPlayers()		=====	Batsman:
print('=====')		=====	Player name: Mahmudullah Jersey No: 30
p2=Player('Mustafizur Rahman','Bowler',90)		Mustafizur Rahman added in team	Player name: Tamim Iqbal Jersey No: 28
p3=Player('Tamim Iqbal',28)		A player cannot be added without a role.	Player name: Mushfiquir Rahim Jersey No: 15
print('=====')		=====	Bowler:
team.addPlayer(p2,p3)		Batsman:	Player name: Mustafizur Rahman Jersey No: 90
print('=====')		Player name: Mahmudullah, Jersey No: 30	Player name: Taskin Ahmed Jersey No: 3
team.showPlayers()		Bowler:	
print('=====')		Player name: Mustafizur Rahman Jersey No: 90	
p3.setRole('Batsman')		=====	
print('=====')		=====	
team.addPlayer(p3)		Tamim Iqbal added in team	
print('=====')		=====	
team.showPlayers()		Batsman:	
print('=====')		Player name: Mahmudullah Jersey No: 30	
p4=Player('Mushfiquir Rahim','Batsman',15)		Player name: Tamim Iqbal Jersey No: 28	
p5=Player('Taskin Ahmed', 'Bowler',3)		Bowler:	
print('=====')		Player name: Mustafizur Rahman Jersey No: 90	

Question-3

Driver Code	Output
<pre>country1= olympicCountry('Jamaica') country2= olympicCountry('USA') player1=AwardedPlayer('Usain Bolt') player2=AwardedPlayer('Shelly-Ann Fraser-Pryce','Silver') player3=AwardedPlayer('Elaine Thompson Herah','Gold') player4=AwardedPlayer('Michael Phelps','Gold') player5=AwardedPlayer('Caeleb Dressel','Gold') print('=====') country1.add_player(player1) print('=====') player1.add_medal('Gold') country1.add_player(player1) country1.add_player(player2) country1.add_player(player3) print('=====') country1.show_medal_info() print('=====') country2.add_player(player4) country2.add_player(player5) print('=====') country2.show_medal_info()</pre>	<pre>===== A player must have a medal. Add the medal first ===== ===== Total medals won by Jamaica is 3 ***** Gold['Usain Bolt','Elaine Thompson Herah'] Silver['Shelly-Ann Fraser-Pryce'] Bronze[] ===== ===== Total medals won by USA is 2 ***** Gold['Michael Phelps','Caeleb Dressel'] Silver[] Bronze[]</pre>

(Mid) Question-1

Driver Code		Output	
sds21 = SDS("2021")	print("6.=====")	School of Data and Sciences 2021! Only	Name: Samiul
print("*****")	s3.print_info()	CSE and MNS department students can	ID: 21305001
sds21.print_info()	print("7.=====")	be added.	Department: MNS
print("1.=====")	s4.print_info()	*****	Courses: ['ENG091', 'MAT092', 'PHY111']
s1 = Student("Maliha",21101001)	print("8.=====")	Number of CSE students: 0	CGPA: 3.01
s1.print_info()	sds21.addStudent(s3,s4)	Number of MNS students: 0	7. =====
print("2.=====")	print("9.=====")	CSE Students {}	Name: Mohsina
sds21.addStudent(s1)	sds21.print_info()	MNS Students {}	ID: 21305023
sds21.print_info()	print("10.=====")	1.=====	Department: MNS
print("3.=====")	s5 = Student("Nayla", 6101288, "BBS")	Name: Maliha	Courses: []
s2 = Student("Subha",21301001,"CSE",	sds21.addStudent(s5)	ID: 21101001	CGPA: 0
["ENG091","MAT092"],3.48)	print("11.=====")	Department: MNS	8.=====
s2.print_info()	sds21.print_info()	Courses: []	9.=====
print("4.=====")		CGPA: 0	Number of CSE students: 1
sds21.addStudent(s2)		2.=====	Number of MNS students: 3
sds21.print_info()		Number of CSE students: 0	CSE Students {21301001: ['Subha', 3.48,
print("5.=====")		Number of MNS students: 1	['ENG091', 'MAT092']]}
s3 = Student("Samiul",21305001,"MNS",		CSE Students {}	MNS Students {21101001: ['Maliha', 0, []],
["ENG091","MAT092","PHY111"],3.01)		MNS Students {21101001: ['Maliha', 0, []]}	21305001: ['Samiul', 3.01, ['ENG091', 'MAT092',
s4 = Student("Mohsina",21305023)		3.=====	'PHY111'], 21305023: ['Mohsina', 0, []]}
		Name: Subha	10.=====
		ID: 21301001	BBS department student cannot be added to SDS
		Department: CSE	11.=====
		Courses: ['ENG091', 'MAT092']	Number of CSE students: 1
		CGPA: 3.48	Number of MNS students: 3
		4.=====	CSE Students {21301001: ['Subha', 3.48,
		Number of CSE students: 1	['ENG091', 'MAT092']]}
		Number of MNS students: 1	MNS Students {21101001: ['Maliha', 0, []], 21305001:
		CSE Students {21301001: ['Subha', 3.48, :	['Samiul', 3.01, ['ENG091', 'MAT092', 'PHY111']],
		['ENG091', 'MAT092']]}	21305023: ['Mohsina', 0, []]}
		MNS Students {21101001: ['Maliha', 0, []]}	
		5.=====	
		6.=====	

Question-2

Driver Code	Output
<pre>dom = Dominos("Domino's Pizza") print("1.=====") p1 = Pizza("Spicy Chicken","Spicy Chicken","Onion","Capsicum") dom.addPizza(p1) p2 = Pizza("Margherita","Mozzarella Cheese") print("2.=====") p3 = Pizza("Beef Kala Bhuna") print("3.=====") p3.setToppings("Curried Beef","Capsicum","Paprika","Onion") print("4.=====") dom.addPizza(p2,p3) p4 = Pizza("Chicken Dominator","Barbecue Chicken,Spicy Chicken,Grilled C hicken") p5 = Pizza("Beefzza","Minced Beef","Beef Pepperoni","Onion","Jalapeno") dom.addPizza(p4,p5) print("5.=====") dom.showMenu() print("6.=====")</pre>	<pre>Domino’s. The Pizza Delivery Experts. 1.===== 2.===== A pizza without toppings cannot be created. 3.===== Toppings added to Beef Kala Bhuna pizza. 4.===== 5.===== Total number of pizzas: 5 Chicken Pizza:2 Name:Spicy Chicken, Toppings: ['Spicy Chicken', 'Onion', 'Capsicum'] Name:Chicken Dominator, Toppings: ['Barbecue Chicken,Spicy Chicken,Grilled Chicken'] Beef Pizza:2 Name:Beef Kala Bhuna, Toppings: ['Curried Beef', 'Capsicum', 'Paprika', 'Onion'] Name:Beefzza, Toppings: ['Minced Beef', 'Beef Pepperoni', 'Onion', 'Jalapeno'] Cheese Pizza: 1 Name:Margherita, Toppings: ['Mozzarella Cheese'] 6.=====</pre>

Question-3

Driver Code		Output	
<pre>argentina = Team("Argentina", 2) print("1.=====") print(argentina.player_list) print("2.=====") argentina.calculateCost() print("3.=====") argentina.printDetail() print("4.=====") messi = Player("Messi") maria = Player("Maria") marcos = Player("Marcos", "Defender") argentina.addPlayer(messi) argentina.addPlayer(maria) argentina.addPlayer(marcos) print("5.=====") argentina.calculateCost() print("6.=====") argentina.printDetail() print("7.=====") print(argentina.player_list) print("8.#####") brazil = Team("Brazil", 5, "Portuguese") print("9.=====") print(brazil.player_list) print("10.=====") brazil.calculateCost() print("11.=====") brazil.printDetail()</pre>	<pre>print("12.=====") silva = Player("Silva", "Defender") neymar = Player("Neymar") brazil.addPlayer(silva) brazil.addPlayer(neymar) print("13.=====") brazil.calculateCost() print("14.=====") brazil.printDetail() print("15.=====") print(brazil.player_list)</pre>	<pre>1.===== No players assigned yet 2.===== No cost yet as there is no players 3.===== Team Name: Argentina World Cup Win: 2 Language: Spanish 4.===== 5.===== Total Cost: 1450 6.===== Team Name: Argentina 'Forward']] World Cup Win: 2 Language: Spanish Forwards: Messi Maria Defenders: Marcos 7.===== [('Messi', 'Forward'), ('Maria', 'Forward'), ('Marcos', 'Defender')] 8.##### 9.===== No players assigned yet 10.===== No cost yet as there is no players 11.===== Team Name: Brazil World Cup Win: 5 Language: Portuguese</pre>	<pre>12.===== 13.===== Total Cost: 950 13.===== Team Name: Brazil World Cup Win: 5 Language: Portuguese Forwards: Neymar Defenders: Silva 15.===== [('Silva', 'Defender'), ('Neymar',</pre>

Question-4

Driver Code	Output
<pre>fareChart = {'SR Travels': 40, 'ENA': 50, 'Pathao': 200, 'Uber': 250, 'Hanif Paribahan': 45, 'Others': 90 } v1 = Vehicle("Dhaka-123", "Bus", 'SR Travels', fareChart) p1 = Passenger("Sam", "Dhaka") print("1.=====") v1.addPassenger(p1) print("2.=====") p1.calculateFare(v1) print("3.=====") p2 = Passenger("John", "Dhaka") p3 = Passenger("Bran") print("4.=====") p3.setDestination("Chittagong") print("5.=====") v1.addPassenger(p2, p3) print("6.=====") p2.calculateFare(v1) print("7.=====") p4 = Passenger("Mark", "Dhaka") p5 = Passenger("Jack", "Khulna") v1.addPassenger(p4, p5) print("8.=====") p4.calculateFare(v1) print("=====") v1.status() v2 = Vehicle("Chittagong-798", "Car", "Uber", fareChart) print("9.=====") v2.addPassenger(p1) v2.addPassenger(p3) print("10.=====") p3.calculateFare(v2)</pre>	<pre>print("11.=====") v2.status() v3 = Vehicle("Khulna-997", "Car", "Obhai", fareChart) v3.addPassenger(p5) print("12.=====") p5.calculateFare(v3) print("13.=====") v3.status()</pre> 1.===== Sam has taken the Bus. 2.===== Calculating fare for Sam 3.===== Destination cannot be empty! 4.===== Destination of Bran is updated to Chittagong 5.===== John has taken the Bus. This is an Inter-City Bus. Bran cannot take the Bus. 6.===== Calculating fare for John 7.===== Mark has taken the Bus. This is an Inter-City Bus. Jack cannot take the Bus. 8.===== Calculating fare for Mark ===== Details of Dhaka-123 Bus of SR Travels: Total Passengers: 3 Passenger name : Sam, Destination: Dhaka, Ticket Price: 40 Passenger name : John, Destination: Dhaka, Ticket Price: 40 Passenger name : Mark, Destination: Dhaka, Ticket Price: 40 9.===== This is an Inter-City Car. Sam cannot take the Car. Bran has taken the Car. 10.===== Calculating fare for Bran 11.===== Details of Chittagong-798 Car of Uber Total Passengers: 1 Passenger name : Bran, Destination: Chittagong, Ticket Price: 250 Jack has taken the Car. 12.===== Calculating fare for Jack 13.===== Details of Khulna-997 Car of Obhai: Total Passengers: 1 Passenger name : Jack, Destination: Khulna, Ticket Price: 90

Question-5

Driver Code	Output
<pre>fareChart = {"Bus":550, "Train":600} p1 = Passenger("Bob", "Bus Ticket",800) p2 = Passenger("Simon", "Train Ticket", 950) p3 = Passenger("David", "Train Ticket", 750) p4 = Passenger("Carol", 1500) print("1=====") tc = TicketCounter() print("2=====") tc.servePassenger(p1,fareChart) print("3=====") tc.servePassenger(p2,fareChart) print("4=====") tc.servePassenger(p3, p4, fareChart) print("5=====") p4.buyTicket("Train Ticket") tc.servePassenger(p4,fareChart) print("6=====") tc.showAllPassenger()</pre>	<pre>Bob is waiting for Bus having 800 taka in pocket. Simon is waiting for Train having 950 taka in pocket. David is waiting for Train having 750 taka in pocket. Carol has 1500 taka in pocket.Please buy a ticket. 1===== This is a general bus and train ticket counter. 2===== Bob has bought Bus ticket. 3===== Simon has bought Train ticket. 4===== David has bought Train ticket. Carol please buy a ticket first. 5===== Carol has bought Train ticket. 6===== Total passengers:4 Bus passengers:1 Passenger Name:Bob Remaining balance:250 Bus passengers:3 Passenger Name:Simon Remaining balance:350 Passenger Name:David Remaining balance:150 Passenger Name:Carol Remaining balance:900</pre>

Question-6

Driver Code

```
fareChart = {'Banani': 2, 'Mohakhali':3, 'Mohammadpur':4, 'New
Market':5, 'Others':6}
v1 = Vehicle("Dhaka-123", "Bus", 3)
p1 = Passenger("Sam", "Banani")
print("1.=====")
v1.addPassenger(p1)
print("2.=====")
p1.calculateFare(v1)
print("3.=====")
p2 = Passenger("John", "Mohakhali")
p3 = Passenger("Bran")
print("4.=====")
p3.setDestination()
print("5.=====")
p3.setDestination("Kakrail")
print("6.=====")
v1.addPassenger(p2)
print("7.=====")
p2.calculateFare(v1)
print("8.=====")
v1.addPassenger(p3)
print("9.=====")
p3.calculateFare(v1)
p4 = Passenger("Mark", "Mohammadpur")
print("10.=====")
p4.setDestination()
print("11.=====")
v1.addPassenger(p4)
v1.addPassenger(Passenger("Jack", "New Market"))
print("12.=====")

v1.details()
v2 = Vehicle("Chittagong798", "Car", 0)
print("13.=====")
v2.addPassenger(p1)
v2.addPassenger(p3)
print("14.=====")
v2.details()
```

Output

```
1.=====
Sam has taken the Bus.
2.=====
Calculating fare for Sam
3.=====
Destination cannot be empty!
4.=====
Destination cannot be empty!
5.=====
Destination of Bran is updated to Kakrail
6.=====
John has taken the Bus.
7.=====
Calculating fare for John
8.=====
Bran has taken the Bus.
9.=====
Calculating fare for Bran
10.=====
Destination cannot be empty!
11.=====
Bus is loaded. Mark cannot take the Bus.
Bus is loaded. Jack cannot take the Bus.
12.=====
Details of Dhaka-123 Bus:
Total Passengers: 3
Passenger name : Sam, Destination: Banani, Fare: 20
Passenger name : John, Destination: Mohakhali, Fare: 30
Passenger name : Bran, Destination: Kakrail, Fare: 60
13.=====
Car is loaded. Sam cannot take the Car.
Car is loaded. Bran cannot take the Car.
14.=====
Details of Chittagong-798 Car:
Total Passengers: 0
```

Question-7

For each main character the cost is 10000
For each supporting character the cost is 5000

Driver Code		Output	
sherlock = Series("Sherlock Holmes", 2010)	print("8.#####")	1.=====	Series Name: Squid Games
print("1.=====")	squid_games = Series("Squid Games", 2021, "Korean")	No characters assigned yet	Release Year: 2021
print(sherlock.character_list)	print("9.=====")	2.=====	Language: Korean
print("2.=====“")	print(squid_games.character_list)	No cost yet as there is no characters	12.=====
sherlock.calculateCost()	print("10.=====")	3.=====	13.=====
print("3.=====")	squid_games.calculateCost()	Series Name: Sherlock Holmes	Total Cost: 15000
sherlock.printDetail()	print("11.=====")	Release Year: 2010	14.=====
print("4.=====")	squid_games.printDetail()	Language: English Series	Name: Squid Games
holmes = Character("Sherlock Holmes", "Main character")	print("12.=====")	4.=====	Release Year: 2021
watson = Character("James Watson", "Main character")		5.=====	Language: Korean
irene = Character("Irene Adler")	seong = Character("Seong", "Main character")	Total Cost: 25000	Main Characters(s):
sherlock.addCharacter(holmes)	kang = Character("Kang")	6.=====	Seong
sherlock.addCharacter(watson)	squid_games.addCharacter(seong)	Series Name: Sherlock Holmes	Supporting Characters(s):
sherlock.addCharacter(irene)	squid_games.addCharacter(kang)	Release Year: 2010	Kang
print("5.=====")	print("13.=====")	Language: English	15.=====
sherlock.calculateCost()	squid_games.calculateCost()	Main Characters(s):	[('Seong', 'Main character'), ('Kang',
print("6.=====")	print("14.=====")	Sherlock Holmes'	'Supporting character')]
sherlock.printDetail()	squid_games.printDetail()	James Watson	
print("7.=====")	print("15.=====")	Supporting Characters(s):	
print(sherlock.character_list)	print(squid_games.character_list)	Irene Adler	
		7.=====	
		[('Sherlock Holmes', 'Main character'), ('James Watson', 'Main character'),	
		('Irene Adler', 'Supporting character')]	
		8.#####	
		9.=====	
		No characters assigned yet	
		10.=====	
		No cost yet as there is no characters	
		11.=====	